

Type-Safe Is Not Error-Free: Typed Decision Models Follow the Option Name, Not the Definition Bound to It

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Abstract

Typed decision models return structured results, but output-type correctness alone does not ensure that decisions follow explicit option definitions. Each option pairs a name with a definition that defines its intended meaning; the name, however, can provide a competing semantic cue. We study this conflict in Jev and two open-weight models by changing only the name-definition mapping, leaving the question, state, and the names and definition texts themselves unchanged. We measure decision flips at the level of the selected definition, rather than the returned name. On 1200 decision tasks with task-specific definitions, decision-flip rates are up to 70.4 pp higher with yes/no names than with the 0/1 control. This gap holds across all 4 binary decision rules. With yes/no names, reassignment also lowers their mean AUC from 93.8% to a below-chance 23.2%. In the binary evaluations, random strings used as option names yield mean flip rates close to those of neutral controls across all three models, with comparable balanced accuracy before reassignment. Together, these results support option-name polarity as a contributor to decision instability beyond reassignment alone. The type-error rate remains 0% throughout, showing that type-correct outputs can still fail to follow explicit option definitions.

normalization. The blog post introducing Jev therefore presents the model’s 0% type-error rate as a structural guarantee rather than an empirical measurement.¹ Type safety restricts which options a model can return, but does not ensure that it selects among them according to their explicit definitions.

Each option pairs a short name with a definition that defines its intended meaning for the task. Names such as yes/no, however, carry conventional meanings that may conflict with these definitions. Just as a cat named “Dog” remains a cat, an option’s intended meaning is defined by its definition, not by the conventional meaning of its name. We therefore ask whether models follow explicit option definitions when option names provide competing semantic cues.

To study this conflict, we keep the question, the state, the option names, and the definition texts fixed, and change only how names are paired with definitions. We track changes in the selected definition rather than in the returned name. Under a binary swap, a model that follows the definitions should preserve its definition-level choice and therefore return a different name. Returning the same name instead selects a different definition. Each input states the name–definition assignments explicitly, so the model need not infer them from demonstrations.

Prior work has shown that language models are sensitive to prompt format, option identifiers, and label words (Sclar et al., 2024; Zheng et al., 2024; Liusie et al., 2023), and that smaller models tend to follow the semantic priors of label words rather than override them (Wei et al., 2023). Recent studies further show that schema-valid outputs can still be semantically incorrect, and that a schema can take precedence over an explicit instruction (Li, 2026; Singh et al., 2026; Usman, 2026; Lin,

1 Introduction

Typed decision models, such as Jev, return decisions rather than free-form text. Given a typed question, a state, and a declared set of options, such a model returns a probability distribution over that set. Here, type safety refers to the guarantee that the model assigns zero probability to any candidate outside the declared option set. In Jev, this property holds by construction because scores for candidates outside the set are masked before

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¹<https://typesafe.ai/blog/introducing-system-one-models-and-jev>

2026). Our work asks a narrower question: when an option’s name and its definition suggest different meanings, which one determines the model’s decision? We study this question on models whose outputs are type-safe by construction, so the observed failures cannot be attributed to invalid outputs.

Under this setting, the effect of polar names goes beyond a drop in accuracy. For yes/no names, re-assignment lowers the AUC of one model below chance, meaning that its scores rank negative instances above positive ones more often than the reverse. Because AUC is independent of the decision threshold, this behavior cannot be corrected by threshold adjustment.

Our main findings are as follows.

1. Option names can override explicit definitions. Polar-name swaps affect decisions across every tested predicate and can reverse score rankings, rather than merely change individual answers.
2. Polarity contributes to decision instability. Familiar polar names, including interface defaults, are most affected, while random strings behave like neutral identifiers with comparable accuracy.
3. Name dependence extends beyond binary decision tasks and a single model family. Neutral renaming changes non-binary decisions, and sensitivity to names recurs across model families, with effect sizes varying with read-out geometry.

2 The scope of the type-safety guarantee

This section introduces the notation for the models we study and states what their type-safety guarantee establishes.

Typed decision models. A typed decision model receives a state s and a question with instructions q and k options $O = (o_1, \dots, o_k)$. The question’s criteria map each option name n_i , such as **no** or **yes**, to a definition d_i of what the option means; we write $o_i = (n_i, d_i)$. The model assigns a score z_i to each option, returns $p = \text{softmax}(z)$ over O , and answers with the most probable option. We study Jev (TypeSafe, n.d.) and two Jev-like models with open weights, Laya (Convai Innovations, n.d.) and Open-Jev (com-kotobalabs, n.d.). Jev supports three question types, Choice, Score and Noul; the setting above is the Choice type, which

is the one we study. Jev is a closed-source model accessed through an API, so we observe only p . Laya is built on ModernBERT-large (Warner et al., 2024) and Open-Jev on DeBERTa-v3-large (He et al., 2023). In both, each option enters the input as `<name>: <definition>` together with the state and the instructions, so the name and the definition are both visible to the model. They differ in how z_i is computed. Laya uses the contextual embedding of a [MASK] marker token placed at the option. Open-Jev uses the average over all tokens of the option’s text, covering both the name and the definition. Section 4.5 discusses this difference as one possible reason why the option name affects the two models to different degrees.

Type safety. Because p is a distribution over O and the answer is its arg max, the answer is always an element of O , for any input and any weights. In the open-weight models this holds by construction, since scores of candidates outside O are masked before normalization. For Jev it is the documented behavior: its documentation states that the model never makes type errors (TypeSafe, n.d.).

What type safety does not cover. Type safety constrains which options can receive probability, but not how probability is distributed among them. This has two consequences for evaluation. First, a 0% type-error rate says nothing about decision quality, since it follows from how the output is constructed rather than from what the model has learned. Second, the guarantee holds equally in every condition of this paper, including those that invert the decision; a type check therefore cannot detect the failure we study. What type safety leaves open is whether the model interprets the options as intended: whether it chooses an option because its definition matches the state, or because of its name.

3 Reassigning names and definitions

On ordinary questions the name and the definition agree, so accuracy cannot tell which one the model follows. We therefore change which definition is bound to which name and keep everything else fixed.

Reassignment. Each binary question has two definitions: d_{no} states that the condition in the instructions does not hold, and d_{yes} that it does (e.g. “No human attention is warranted.” / “A human should inspect this run.”). For option names (n_0, n_1) , the *aligned* arm binds n_0 to d_{no} and n_1 to d_{yes} , as in

the data; the *reassigned* arm binds n_0 to d_{yes} and n_1 to d_{no} . Nothing else changes. The new binding is written out in the input, and the gold answer follows the definition. A model that reads the definitions is unaffected by the reassignment; a model that reads the names inverts its answer.

Option names. We use five polar pairs: *yes/no* and *false/true*, which Laya emitted during training, and *absent/present*, *negative/positive* and *rejected/accepted*, which it did not. The neutral pairs *0/1* and *A/B* are the control: for them the reassignment only reorders the definitions, so any change measures the operation itself. We report every effect as a difference against them. Sections 4.3 and 4.4 extend the design to random-string names and to more than two options.

Data. We use 1200 binary questions from Typed Decisions (LocalLLaMA, n.d.), covering four decision tasks with 300 questions each: invoice reconciliation, agent-trace triage, security-alert classification and customer-service escalation. Each question has its own task-specific definitions, and 57.5% of the gold answers are yes. Another 600 questions in the same data use a generic wording whose definitions begin with the words “no” and “yes” themselves; there the name and the definition are confounded, so we exclude them from the main results and report them only as an upper bound (Appendix C). Results for Open-Jev, and the results for Laya in Table 4, use all 1800 questions. The 683 questions with more than two options (Section 4.4) come from the test split of the Open-Jev data (ZefanCai, n.d.).

Metrics. We measure the effect of the reassignment with the flip rate, the fraction of questions on which the model chooses a different definition in the two arms; it needs no gold labels. We report it as a difference against the neutral controls, with 95% intervals from 2,000 bootstrap resamples over states. Accuracy is measured in both arms, as balanced accuracy at a threshold of 0.5 and as AUC. The aligned arm shows how well the model answers the questions to begin with (per decision task for Laya in Appendix A), and AUC, which does not depend on the threshold, separates a lost ranking from a shifted decision boundary. Jev is not deterministic, so for it we also measure a test-retest floor: the flip rate between two identical runs of the aligned arm.

4 Results

All experiments apply the reassignment of Section 3, and the type-error rate is 0% in every condition. Sections 4.1 and 4.2 show that option names override the definitions, Section 4.3 that polarity contributes to this effect, and Sections 4.4 and 4.5 that it extends to non-binary decisions and to other model families.

4.1 Polar names override the definitions

Table 1 shows the result on Laya. When the definitions of *yes/no* are reassigned, 76.9% of the decisions flip, compared with 6.5% for *0/1* and 6.0% for *A/B*. The flip rate for *yes/no* is thus 70.4 pp higher than for the *0/1* control (95% CI [67.6, 73.1]) and 70.9 pp higher than for *A/B* (95% CI [68.2, 73.5]). For every polar pair it is at least 39.3 pp higher than for *0/1*. Balanced accuracy for *yes/no* falls from 87.2% to 28.4%, whereas for *0/1* it stays at 83.7% and 83.2%.

The gap holds across all four decision tasks (Table 2). For *yes/no* the flip rate is at least 56.7% in every decision task, at least 7.4 times that of *0/1*, which never exceeds 11.3%. The other polar pairs vary more across decision tasks, from 12.3% to 92.0%; *negative/positive*, for example, flips 90.3% of the invoice decisions but 13.0% of the security decisions.

4.2 The ranking is reversed

A drop in balanced accuracy can mean that the model has become uncertain, or that it ranks the decisions in the wrong direction. AUC distinguishes the two (Figure 1a). For *yes/no*, reassignment lowers Laya’s AUC from 93.8% to 23.2%, below chance: the scores still separate the two classes, but in the opposite direction. Because AUC does not depend on the decision threshold, this error cannot be corrected by adjusting the threshold.

Flip rate and AUC can therefore disagree. For *negative/positive*, 48.3% of the decisions flip, yet AUC falls only from 82.6% to 75.8%: reassignment changes many decisions but leaves most of the ranking intact. Among the polar pairs, only *yes/no* falls below chance.

4.3 Polarity, amplified by familiarity

The five polar pairs differ in whether Laya emitted them during training (Section 3), which separates two contributions (Table 3). Moving from neutral names to the three polar pairs that Laya never emit-

option names	in voc.	balanced acc.		AUC		flip	difference from 0/1 (pp, 95% CI)
		aligned	reassigned	aligned	reassigned		
yes/no	✓	87.2%	28.4%	93.8%	23.2%	76.92%	+70.42 [+67.58, +73.08]
false/true	✓	90.1%	48.6%	96.2%	57.7%	49.67%	+43.17 [+40.25, +46.08]
absent/present		85.9%	53.5%	94.5%	54.3%	47.75%	+41.25 [+38.17, +44.17]
negative/positive		75.3%	67.3%	82.6%	75.8%	48.25%	+41.75 [+38.75, +44.83]
rejected/accepted		79.1%	66.2%	90.8%	63.6%	45.75%	+39.25 [+36.33, +42.25]
0/1		83.7%	83.2%	94.2%	92.4%	6.50%	
A/B		84.2%	82.2%	94.2%	94.1%	6.00%	

Table 1: Laya on the 1200 questions. Reassignment changes only which definition is bound to which option name, and the gold answer follows the definition. Polar names (top) lose the decision; neutral names (bottom) do not. *in voc.* marks the two pairs Laya emitted during training. AUC below 50% means the ranking is reversed.

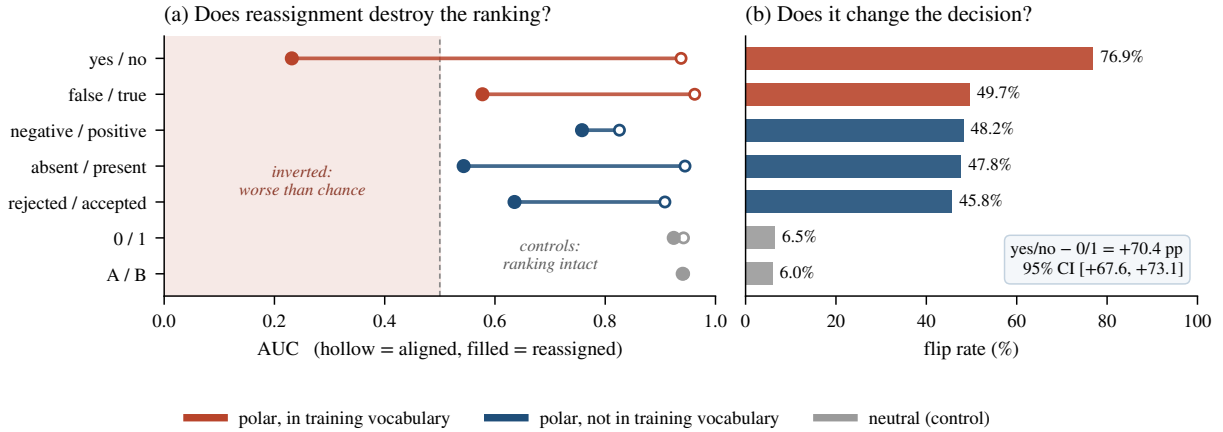


Figure 1: Laya. (a) AUC in the aligned (hollow) and reassigned (filled) arms. The polar pairs fall below chance (shaded): the ranking is reversed, not lost. (b) Flip rate for the same rows. The neutral pairs undergo the same reassignment.

option names	invoice	agent	security	customer	min
yes/no	92.7	83.3	75.0	56.7	56.7
false/true	63.0	39.3	29.0	67.3	29.0
absent/present	56.0	76.7	39.7	18.7	18.7
negative/positive	90.3	31.7	13.0	58.0	13.0
rejected/accepted	92.0	37.0	12.3	41.7	12.3
0/1	2.7	11.3	6.3	5.7	2.7

Table 2: Flip rate (%) of Laya by decision task.

option names	flip	range
neutral (0/1, A/B)	6.25%	2.7–11.3%
polar, unseen in voc. (3 pairs)	47.25%	12.3–92.0%
polar, in voc. (2 pairs)	63.29%	29.0–92.7%
polarity, over neutral		+41.00 pp
training-vocabulary familiarity, on top		+16.04 pp

Table 3: Flip rate of Laya by name class. Ranges are over the four decision tasks.

ted raises the flip rate by 41.0 pp; moving to the two pairs it did emit adds a further 16.0 pp. Polarity accounts for the larger part, and it acts on names that Laya never produced during training.

To test whether the names’ content matters beyond polarity, we name each option with a random five-character string of letters and digits, such as `xg6a6/e97ce`. No string is a word, neither string of a pair is a prefix of the other, and the two have no ordinal or alphabetical relation. We draw 15 such pairs (Appendix D) and run all of them on Laya and Open-Jev, and the first five on Jev. In

all three models, random strings behave like the neutral names (Table 4). Their flip rate differs from that of 0/1 and A/B by at most 3.2 pp: 6.9% against 7.9% on Laya, 11.7% against 14.8% on Open-Jev, and 2.0% against 1.9% on Jev. On the same questions, the two familiar polar pairs flip 32.2% to 70.7% of the decisions. The low flip rate does not come from a loss of accuracy: balanced accuracy before reassignment differs from that with neutral names by at most 3.3 pp.

Individual strings vary more on Open-Jev, from 2.0% to 31.6% across the 15 draws, than on Laya

model	polar		no polarity	
	in voc.	unseen in voc.	neutral	random
<i>flip rate</i>				
Laya	70.72%	50.22%	7.94%	6.86%
Open-Jev	33.75%	18.57%	14.83%	11.67%
Jev	32.21%	11.06%	1.88%	2.02%
<i>balanced accuracy, aligned arm</i>				
Laya	89.6%	79.4%	87.1%	86.2%
Open-Jev	60.1%	59.8%	63.9%	60.6%
Jev	71.1%	70.9%	71.5%	71.5%

Table 4: Random strings against the other name classes. Random-string values are means over 15 draws (five for Jev); the other columns are means over the pairs of Table 1.

option names	acc.	changed
original	56.4%	—
letters <i>A, B, C</i>	27.8%	52.42%
numbers <i>option 1</i>	25.9%	55.78%
reassigned one step	15.5%	79.65%

Table 5: Laya on questions with more than two options. The definitions are unchanged, and the gold answer follows the definition.

(3.2%–13.3%) and Jev (1.7%–2.4%) (Figure 2).

4.4 Beyond binary decisions

We run Laya on 683 questions with three to sixteen options from the test split of the Open-Jev data. The definitions stay unchanged and only the names change, in three ways: neutral letters (*A, B, C, ...*), numbers (*option 1, option 2, ...*), and the original names reassigned one step, so that each name is attached to the definition of its neighbour. The gold answer follows the definition.

Even neutral letters change 52.4% of the decisions and lower accuracy from 56.4% to 27.8%; numbers change 55.8%. Reassigning the original names changes 79.7% and lowers accuracy to 15.5% (Table 5). Appendix E gives an example.

4.5 Other model families

The effect recurs in Jev and Open-Jev (Table 6). In all three models, neutral names flip the fewest decisions, the three polar pairs that Laya never emitted more, and the two it emitted the most. On Jev, evaluated on the same 1,200 questions as Laya, *yes/no* flips 32.5% of the decisions, 30.4 pp more than *0/1* (95% CI [27.6, 33.3]); balanced accuracy falls from 71.3% to 51.6%, and AUC from 81.5% to 58.1%. Because Jev is not deterministic, we run the aligned arm twice on 300 questions. At

option names	Laya	Open-Jev	Jev
<i>yes/no</i>	76.92%	19.50%	32.50%
<i>false/true</i>	49.67%	47.83%	31.92%
<i>absent/present</i>	47.75%	24.06%	15.17%
<i>negative/positive</i>	48.25%	23.44%	4.17%
<i>rejected/accepted</i>	45.75%	8.17%	13.83%
<i>0/1</i>	6.50%	3.61%	2.08%
<i>A/B</i>	6.00%	26.00%	1.67%
polar, in voc. (2 pairs)	63.29%	33.66%	32.21%
polar, unseen in voc. (3 pairs)	47.25%	18.56%	11.06%
neutral (<i>0/1, A/B</i>)	6.25%	14.80%	1.88%
questions	1200	1800	1200

Table 6: Flip rate (%) by name pair (top) and name class (bottom) for the three models. The ordering of the name classes holds in all three; the magnitudes differ.

most 1.33% of the decisions change between the two runs, so the flip rate for *yes/no* is 24 times this floor, while the neutral pairs stay close to it. On Open-Jev, *yes/no* flips 19.5% of the decisions, 15.9 pp more than *0/1*.

The size of the effect differs across models: for *yes/no*, Laya flips 2.4 times as often as Jev and 4.1 times as often as Open-Jev. One possible reason for the difference between the two open-weight models is how they score an option (Section 2). Laya scores a marker token placed at the option, whereas Open-Jev averages over all tokens of the option’s text, so the one-token name is averaged with a much longer definition.

5 Related work

Krishna Kumar (2025) is the closest result: across eight tasks and eight 1–12B decoders, inverted in-context demonstrations produce a semantic override rate of exactly zero. We find the same anchoring in typed decision models, with three differences. The reassignment here is written out in the input rather than implied by demonstrations, so the model does not have to infer it. The output is a softmax restricted to the declared options, so the failure is invisible to a type check. And because the gold answer follows the definition, following the name shows up as an AUC of 23.2%: the ranking is reversed, not merely unchanged. Le (2026) treats schema *keys* as an instruction channel under constrained decoding in decoder LLMs and reports accuracy deltas; we vary option *names* in typed decision models, use neutral names as a control for the reassignment itself, and observe a reversed ranking rather than a drop in accuracy. Lim et al. (2026) names our failure mode from the other side

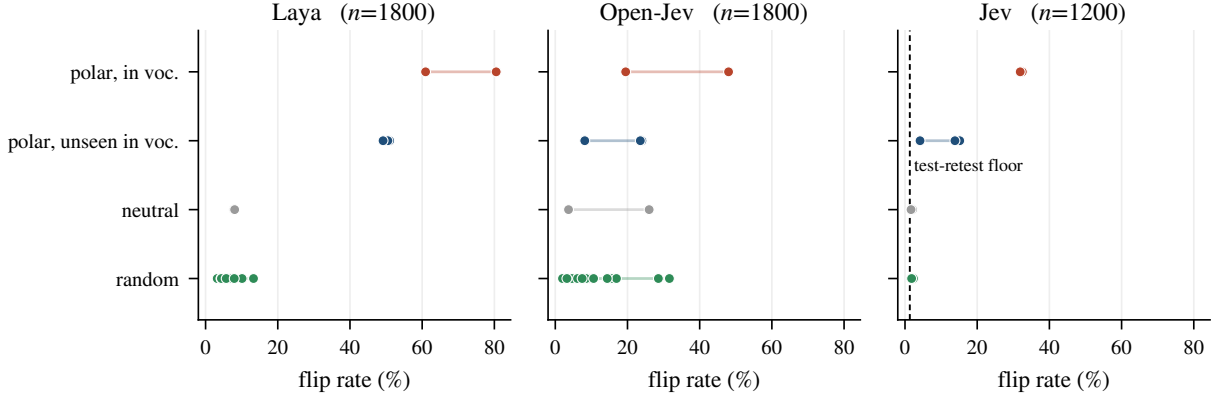


Figure 2: Flip rate by name class, one dot per name pair: 15 random-string draws for Laya and Open-Jev, five for Jev. The dashed line is Jev’s test-retest floor (Section 4.5).

— safety judging as a rubric-following problem, with judges brittle under rubric variation — and proposes the curriculum fix we do not attempt: they vary the rubric while holding names fixed, we hold the definitions fixed and vary which name each is bound to. Work on sensitivity to label words and prompt format (Sclar et al., 2024; Zheng et al., 2024; Liusie et al., 2023; Wei et al., 2023) predicts the direction of our effect; Badhe et al. (2026) analyses the renormalization step these models perform. An independent evaluation harness reports option-order sensitivity in Laya (Vignesh Labs, 2026), a complementary axis to the one studied here.

6 Discussion

What to report next to a type-error rate. A 0% type-error rate follows from how the output is constructed (Section 2), and presenting it as a reliability number invites exactly the inference it cannot support. It should be accompanied by a name-invariance number. The flip rate under re-assignment, measured against neutral names, is a practical candidate: it requires no gold labels, costs two extra forward passes per question, and on Laya it reveals a flip rate of 76.9% for *yes/no* that accuracy with the original names does not show.

For practitioners. Two mitigations follow directly. Use neutral option names and carry the meaning in the definitions, which on Laya gives a flip rate of 6.5% instead of 76.9%; or train the model to follow the definitions by randomizing option names, which is cheap for a model of Laya’s size. Curriculum-based rubric-following (Lim et al., 2026), option-ID debiasing (Zheng et al., 2024) and word-bias correction (Liusie et al., 2023) are available mitigations we do not evaluate

here.

7 Limitations

We study two open-weight models, which differ in how they score an option, on English questions only; whether the explanation by scoring (Section 4.5) holds for other architectures is untested. Open-Jev also answers less accurately before the re-assignment: its balanced accuracy is 59.8%–63.9% across name classes, against 79.4%–89.6% for Laya (Table 4), so its lower flip rate may also reflect a preference for one definition regardless of the state. Jev is a closed-source model accessed through an API at one point in time: we observe only its returned distribution, it is not deterministic — hence the test-retest floor of Section 4.5 — and the model served under that name may change. Name pairs differ in how far they preserve meaning, which is why Section 4.1 reports the minimum over decision tasks rather than the mean; a human annotation of meaning preservation across name pairs would sharpen the middle rows of Table 3 and is the natural next step. On the questions with more than two options, accuracy with the original names is only 56.4%, so we treat that result as directional. We report the vulnerability and two mitigations but evaluate neither.

8 Conclusion

Type safety guarantees that a typed decision model returns an option from the declared set, but it does not guarantee that the model interprets those options according to their explicit definitions. Holding the question, state, option names, and definition texts fixed, we find that changing only which definition is bound to which name can substantially alter

model decisions. For **yes/no**, the flip rate under this reassignment is 70.4 pp higher than under the neutral 0/1 control, and AUC falls from 93.8% to 23.2%. These results show that structurally valid decisions can remain strongly dependent on the semantic cues carried by option names. Type safety guarantees where a decision can land, but not what the options mean to the model.

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A Accuracy of the aligned arm

B Flip rate by decision task

Figure 3 plots Table 2; Figure 4 plots Table 3, with the range over decision tasks as whiskers.

decision task	instructions	<i>n</i>	bal. acc.	AUC
invoice	<i>“The invoice reconciles with the purchase order and the recorded delivery.”</i>	300	87.8%	97.1%
agent	<i>“This trace requires human review.”</i>	300	87.4%	94.7%
security	<i>“This alert reflects genuinely malicious or unauthorised activity.”</i>	300	93.1%	98.7%
customer	<i>“This conversation requires a human agent rather than automated handling.”</i>	300	78.6%	91.4%

Table 7: Laya, aligned arm, yes/no, by decision task.

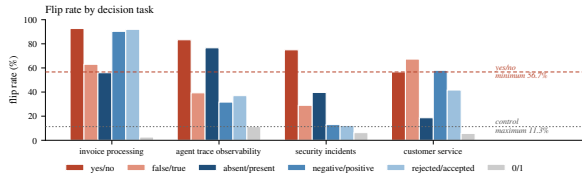


Figure 3: Flip rate of Laya by decision task and name pair.

C Questions with generic definitions

On the 600 questions excluded in Section 3, whose definitions begin with the words “no” and “yes”, reassignment flips 87.7% of the decisions for yes/no, 77.2 pp more than for 0/1 (95% CI [73.0, 81.3]). AUC falls to 8.6% for yes/no and 3.0% for false/true. Because the name and the definition are confounded on these questions, we report this only as an upper bound.

D Random-string names

Table 8 lists the 15 pairs of Section 4.3. They come from a seeded generator whose acceptance rule depends only on the names already accepted, so drawing more pairs does not change the earlier ones. The first row lists the five pairs run on Jev.

xg6a6/e97ce	30d6n/j87g8	hh0t5/blj8b	94wwf/dv7vq	6nfxk/7qnyc
gc46d/wgh8i	18gc6/ldqj2	fbg13/cz922	2cdod/lh8uo	yn6by/syji6
k63g1/za2ml	mq5h1/nvoz2	v5mai/paw1t	u5r4o/rw9lp	0se36/4r2qe

Table 8: The 15 random-string name pairs.

E An example with more than two options

The question below is from the test split of the Open-Jev data (Section 4.4). The instructions ask “What exact integer is stored in result after this program finishes?”, and the state is the program

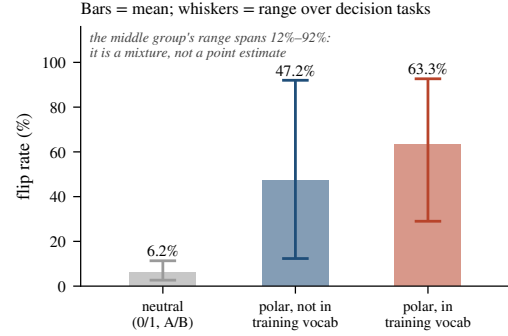


Figure 4: Flip rate of Laya by name class. Bars are means over pairs; whiskers span decision tasks.

```
x = 8
for item in [6, 9, -9, 1]:
    if item > 1:
        x += item
    else:
        x -= 5
result = x * 0
```

The answer is 0. Table 9 gives Laya’s probabilities for the four definitions. With the original names, Laya chooses “The exact integer result is 0” (50.4%). When the names are reassigned one step, the name 0 is attached to “The exact integer result is -17”, and Laya chooses that option (41.2%); the correct definition, now named 5, receives 19.9%.

definition	original		reassigned	
	name	<i>p</i> (%)	name	<i>p</i> (%)
The exact integer result is 5	5	19.4	-3	18.3
The exact integer result is -3	-3	15.6	-17	20.5
The exact integer result is -17	-17	14.5	0	41.2
The exact integer result is 0	0	50.4	5	19.9

Table 9: Laya on one question with four options. Each row is one definition; the name columns give the name attached to it in each arm.